

SULIT



## SECP2523 DATABASE

**SESSION 2024/2025 - SEMESTER 1**

### **ALTERNATIVE ASSESSMENT REPORT: PHASE 2**

|                            |                                             |
|----------------------------|---------------------------------------------|
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| <b>Year / Program</b>      | 2 /SECPH                                    |
| <b>Section</b>             | 2                                           |
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| <b>Case Study</b>          | MEMBER REGISTRATION                         |
| <b>System Name</b>         | Koperasi KADA All-in-one Financial Platform |
| <b>Group Name</b>          | Tech EcoPioneer                             |

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## SECTION C (SQL AND INTERFACE IMPLEMENTATION)

### 3.1 SQL STATEMENT (DDL & DML)

1. Create the Member table:

```
CREATE TABLE Member (  
    memberID INT PRIMARY KEY,  
    memberName VARCHAR(255) NOT NULL,  
    memberIC VARCHAR(22) NOT NULL UNIQUE,  
    memberGender ENUM('Male', 'Female') NOT NULL,  
    memberReligion VARCHAR(50),  
    memberRace VARCHAR(50),  
    memberMarital_status ENUM('Single', 'Married', 'Divorced', 'Widowed') NOT NULL,  
    memberPfNo INT NOT NULL UNIQUE,  
    memberSalary DECIMAL(10,2),  
    memberPosition VARCHAR(100),  
    memberGrade VARCHAR(50),  
    memberStatus ENUM('Pending', 'Lulus', 'Tolak') NOT NULL,  
    applicationDate TIMESTAMP NOT NULL,  
    approvalDate TIMESTAMP  
);
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0251 seconds.)

```
CREATE TABLE Member ( memberID INT PRIMARY KEY, memberName VARCHAR(255) NOT NULL, memberIC VARCHAR(22) NOT NULL UNIQUE, memberGender ENUM('Male', 'Female') NOT NULL, memberReligion  
VARCHAR(50), memberRace VARCHAR(50), memberMarital_status ENUM('Single', 'Married', 'Divorced', 'Widowed') NOT NULL, memberPfNo INT NOT NULL UNIQUE, memberSalary DECIMAL(10,2),  
memberPosition VARCHAR(100), memberGrade VARCHAR(50), memberStatus ENUM('Pending', 'Lulus', 'Tolak') NOT NULL, applicationDate TIMESTAMP NOT NULL, approvalDate TIMESTAMP );
```

2. Create the Validation table:

```
CREATE TABLE Validation (  
    memberID INT,  
    staffID INT,  
    validationDate TIMESTAMP NOT NULL,  
    validationStatus ENUM('Pending', 'Lulus', 'Tolak') NOT NULL,  
    PRIMARY KEY (memberID, staffID),  
    FOREIGN KEY (memberID) REFERENCES Member(memberID),  
    FOREIGN KEY (staffID) REFERENCES Staff(staffID)  
);
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0254 seconds.)

```
CREATE TABLE Validation ( memberID INT, staffID INT, validationDate TIMESTAMP NOT NULL, validationStatus ENUM('Pending', 'Lulus', 'Tolak') NOT NULL, PRIMARY KEY (memberID, staffID), FOREIGN KEY (memberID) REFERENCES Member(memberID), FOREIGN KEY (staffID) REFERENCES Staff(staffID) );
```

### 3. Insert a new staff

```
INSERT INTO Staff (staffID, staffIC, staffGender, staffPfNo, staffPosition, staffGrade)
```

```
VALUES (101, '801201045678', 'Male', 12345, 'Manager', 'Grade A');
```

✓ 1 row inserted. (Query took 0.0042 seconds.)

```
INSERT INTO Staff (staffID, staffIC, staffGender, staffPfNo, staffPosition, staffGrade) VALUES (101, '801201045678', 'Male', 12345, 'Manager', 'Grade A');
```

### 4. Insert new members

```
INSERT INTO Member
```

```
(memberID,memberName,memberIC,memberGender,memberReligion,memberRace,memberMarital_status,memberPfNo,memberSalary,memberPosition,memberGrade,memberStatus,applicationDate, approvalDate)
```

```
VALUES
```

```
(1, 'Wee Leong Chuan', '990000000000', 'Male', 'Christianity', 'Caucasian', 'Single', 12345, 2000.00, 'Pengguna', 'Grade A', 'Pending', CURRENT_TIMESTAMP, NULL),
```

```
(2, 'Teo Kah Wee', '880011010129', 'Male', 'Buddha', 'Cina', 'Married', 54321, 12000.00, 'Doctor', 'Grade B', 'Pending', CURRENT_TIMESTAMP, NULL),
```

```
(3, 'Muhammad ZJ', '040000000000', 'Male', 'Islam', 'Malay', 'Divorced', 67890, 9000.00, 'IT', 'Grade C', 'Lulus', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
```

✓ 3 rows inserted. (Query took 0.0053 seconds.)

```
INSERT INTO Member (memberID,memberName,memberIC,memberGender,memberReligion,memberRace,memberMarital_status,memberPfNo,memberSalary,memberPosition,memberGrade,memberStatus,applicationDate, approvalDate) VALUES (1, 'Wee Leong Chuan', '990000000000', 'Male', 'Christianity', 'Caucasian', 'Single', 12345, 2000.00, 'Pengguna', 'Grade A', 'Pending', CURRENT_TIMESTAMP, NULL), (2, 'Teo Kah Wee', '880011010129', 'Male', 'Buddha', 'Cina', 'Married', 54321, 12000.00, 'Doctor', 'Grade B', 'Pending', CURRENT_TIMESTAMP, NULL), (3, 'Muhammad ZJ', '040000000000', 'Male', 'Islam', 'Malay', 'Divorced', 67890, 9000.00, 'IT', 'Grade C', 'Lulus', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
```

### 5. Insert a Validation Record

```
INSERT INTO Validation (memberID, staffID, validationDate, validationStatus)
```

```
VALUES (1, 101, CURRENT_TIMESTAMP, 'Pending');
```

✓ 1 row inserted. (Query took 0.0131 seconds.)

```
INSERT INTO Validation (memberID, staffID, validationDate, validationStatus) VALUES (1, 101, CURRENT_TIMESTAMP, 'Pending');
```

### 6. Update a Member's Status After Validation

```
UPDATE Member
```

SET memberStatus = 'Lulus', approvalDate = CURRENT\_TIMESTAMP

WHERE memberID = 1;

✓ 1 row affected. (Query took 0.0131 seconds.)

UPDATE Member SET memberStatus = 'Lulus', approvalDate = CURRENT\_TIMESTAMP WHERE memberID = 1;

## 7. Retrieve Validation Details for Each Member

SELECT m.memberID,m.memberName,v.validationDate,v.validationStatus

FROM Member m

JOIN Validation v

ON m.memberID = v.memberID;

✓ Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

SELECT m.memberID,m.memberName,v.validationDate,v.validationStatus FROM Member m JOIN Validation v ON m.memberID = v.memberID;

☐ Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

☐ Show all | Number of rows: 25 ▾ | Filter rows: Search this table

Extra options

| memberID | memberName      | validationDate      | validationStatus |
|----------|-----------------|---------------------|------------------|
| 1        | Wee Leong Chuan | 2025-01-23 11:33:59 | Pending          |

## 8. Retrieve Staff Who Validated Each Member

SELECT s.staffID,s.staffPosition,m.memberName,v.validationDate

FROM Staff s

JOIN Validation v

ON s.staffID = v.staffID

JOIN Member m

ON v.memberID = m.memberID;

✓ Showing rows 0 - 0 (1 total, Query took 0.0008 seconds.)

SELECT s.staffID,s.staffPosition,m.memberName,v.validationDate FROM Staff s JOIN Validation v ON s.staffID = v.staffID JOIN Member m ON v.memberID = m.memberID;

☐ Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

☐ Show all | Number of rows: 25 ▾ | Filter rows: Search this table

Extra options

| staffID | staffPosition | memberName      | validationDate      |
|---------|---------------|-----------------|---------------------|
| 101     | Manager       | Wee Leong Chuan | 2025-01-23 11:33:59 |

## 9. Retrieve Members with Status 'Lulus'

```
SELECT
m.memberID,m.memberName,m.memberIC,m.memberStatus,v.validationDate,v.validationStatus

FROM Member m

JOIN Validation v

ON m.memberID = v.memberID

WHERE m.memberStatus = 'Lulus';
```

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
SELECT m.memberID,m.memberName,m.memberIC,m.memberStatus,v.validationDate,v.validationStatus FROM Member m JOIN Validation v ON m.memberID = v.memberID WHERE m.memberStatus = 'Lulus';
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

| memberID | memberName      | memberIC     | memberStatus | validationDate      | validationStatus |
|----------|-----------------|--------------|--------------|---------------------|------------------|
| 1        | Wee Leong Chuan | 990000000000 | Lulus        | 2025-01-23 11:39:15 | Lulus            |
| 3        | Muhammad ZJ     | 040000000000 | Lulus        | 2025-01-23 11:40:32 | Lulus            |

## 10. Retrieve Members with Status 'Pending'

```
SELECT memberID, memberName, memberIC, memberGender, memberPosition,
memberSalary, memberStatus, applicationDate

FROM Member

WHERE memberStatus = 'Pending';
```

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
SELECT memberID, memberName, memberIC, memberGender, memberPosition, memberSalary, memberStatus, applicationDate FROM Member WHERE memberStatus = 'Pending';
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

| memberID | memberName  | memberIC     | memberGender | memberPosition | memberSalary | memberStatus | applicationDate     |
|----------|-------------|--------------|--------------|----------------|--------------|--------------|---------------------|
| 2        | Teo Kah Wee | 880011010129 | Male         | Doctor         | 12000.00     | Pending      | 2025-01-23 11:30:59 |

## 3.2 INTERFACE IMPLEMENTATION

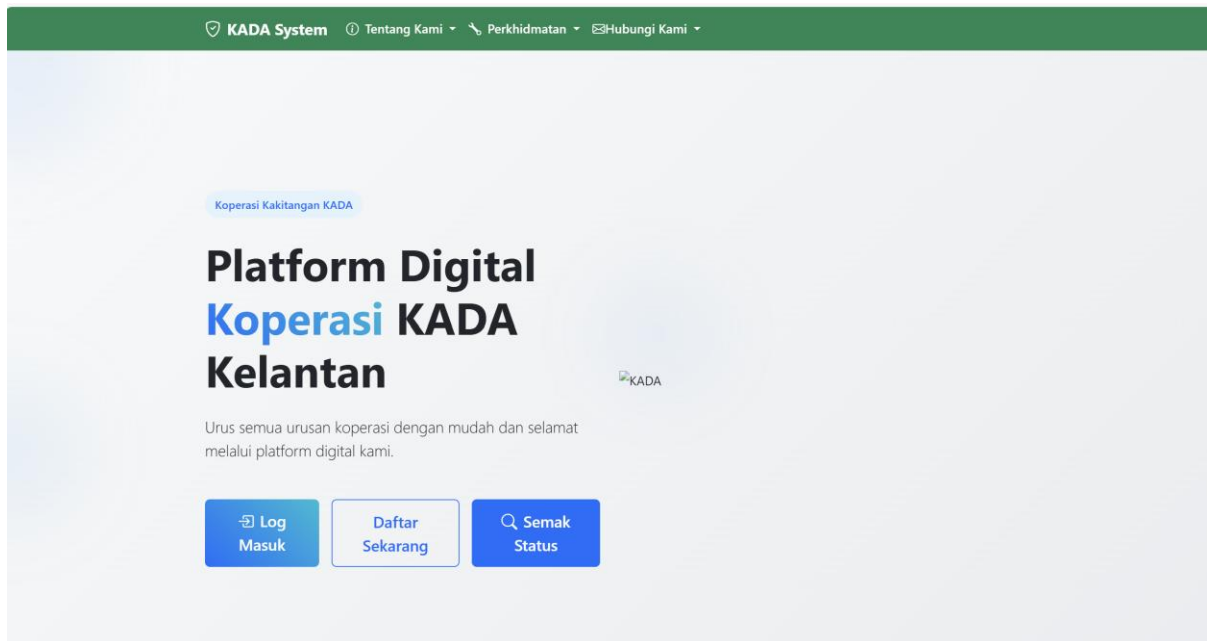


Figure 3.2.1

Figure 3.2.1 shows the landing page of our system, new users can click the button 'Daftar Sekarang' to register as KADA's member. Besides, after submitting the application, they can check the status of their application by clicking the 'Semak Status' button.

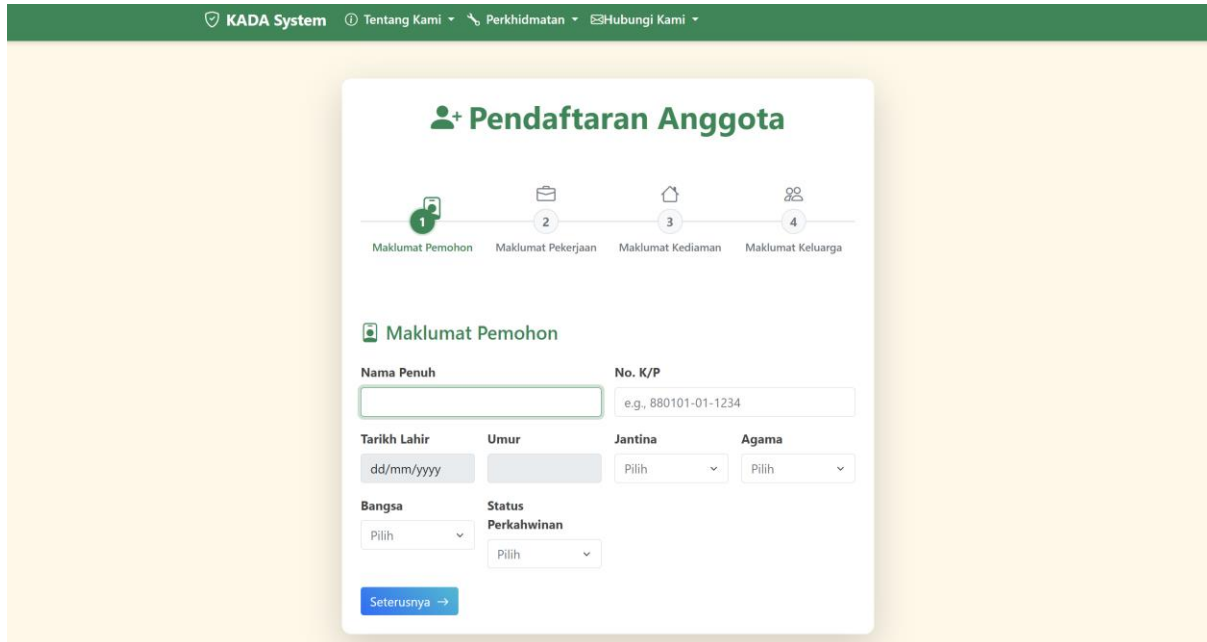


Figure 3.2.2

KADA System
Tentang Kami
Perkhidmatan
Hubungi Kami

## Pendaftaran Anggota

1
Maklumat Pemohon
2
Maklumat Pekerjaan
3
Maklumat Kediaman
4
Maklumat Keluarga

### Maklumat Pekerjaan Pemohon

Gaji Bulanan (RM)

Jawatan

Gred

Pilih gred...

Sebelumnya
Seterusnya

Figure 3.2.3

KADA System
Tentang Kami
Perkhidmatan
Hubungi Kami

## Pendaftaran Anggota

1
Maklumat Pemohon
2
Maklumat Pekerjaan
3
Maklumat Kediaman
4
Maklumat Keluarga

### Maklumat Kediaman

Alamat Rumah

Poskod

Negeri

Pilih Negeri

No. Telefon Bimbit

No. Telefon Rumah

### Alamat

Alamat Pejabat

Poskod

Negeri

Pilih Negeri

No. Telefon Pejabat

No. Fax

Sebelumnya
Seterusnya

Figure 3.2.4

KADA System
Tentang Kami
Perkhidmatan
Hubungi Kami

## Pendaftaran Anggota

1
Maklumat Pemohon
2
Maklumat Pekerjaan
3
Maklumat Kediaman
4
Maklumat Keluarga

### Maklumat Keluarga dan Pewaris

Hubungan Keluarga

Nama

No. K/P atau No. Surat Beranak

Pilih

Tambah Ahli

e.g., 880101-01-1234

Sebelumnya
Hantar



Figure 3.2.5

Figure 3.2.2, 3.2.3, 3.2.4, and 3.2.5 shows the application form for member registration after clicking the ‘Daftar Sekarang’ button.

**Semak Status Permohonan**

No. Kad Pengenalan  Nama Penuh

ATAU

Nombor Rujukan

Contoh: REF202501160001

Sila masukkan nombor rujukan yang diberikan semasa pendaftaran.

[Semak Status](#)

Figure 3.2.6

Figure 3.2.6 shows the page for users to check their application’s status by entering their name and IC or reference number of the application form after clicking the ‘Semak Status’ button.

**Senarai Ahli**

9 Jumlah Ahli | 2 Pending | 6 Ahli Aktif | 1 Ditolak

Cari ahli...

| No. | Nama            | No. K/P        | Jantina | Jawatan     | Gaji         | Status  | Tindakan |
|-----|-----------------|----------------|---------|-------------|--------------|---------|----------|
| 1   | WEE LEONG CHUAN | 990000-00-0000 | Male    | Pengguna    | RM 2,000.00  | Ahli    |          |
| 2   | TAN YEE BAL     | 001020-10-1203 | Female  | PENSYARAH   | RM 1,000.00  | Ahli    |          |
| 3   | TEST            | 011111-11-1111 | Male    | 124         | RM 24,444.00 | Tolak   |          |
| 4   | MUHAMMAD ZJ     | 040000-00-0000 | Male    | IT          | RM 9,000.00  | Ahli    |          |
| 5   | A               | 777777-77-7777 | Male    | Pelajar UTM | RM 4,444.00  | Ahli    |          |
| 6   | LIM ZI MING     | 888888-00-0000 | Female  | IT          | RM 8,000.00  | Ahli    |          |
| 7   | CHEW CX         | 222222-22-2222 | Male    | Free Rider  | RM 300.00    | Ahli    |          |
| 8   | TEST            | 111111-11-1111 | Female  | 12          | RM 1,212.00  | Pending |          |
| 9   | TEO KAH WEE     | 880011-01-0129 | Male    | Doctor      | RM 12,000.00 | Pending |          |

Menunjukkan 1-9 daripada 9 rekod

Figure 3.2.7

Figure 3.2.7 shows the page for staff to validate the members’ application.

| <div>KADA System</div> <div>Tentang Kami</div> <div>Perkhidmatan</div> <div>Hubungi Kami</div> <div>Welcome, dhesh</div> <div>Keluar</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                |         |         |              |         |          |     |      |         |         |         |      |        |          |   |      |                |        |    |             |         |  |   |             |                |      |        |              |         |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|---------|---------|--------------|---------|----------|-----|------|---------|---------|---------|------|--------|----------|---|------|----------------|--------|----|-------------|---------|--|---|-------------|----------------|------|--------|--------------|---------|--|
| <div>Senarai Ahli</div> <div>Export</div> <div> <div>9 Jumlah Ahli</div> <div>2 Pending</div> <div>6 Ahli Aktif</div> <div>1 Ditolak</div> </div> <div> <div>Cari ahli...</div> <div>Pending</div> </div> <table> <tr> <th>No.</th><th>Nama</th><th>No. K/P</th><th>Jantina</th><th>Jawatan</th><th>Gaji</th><th>Status</th><th>Tindakan</th></tr> <tr> <td>1</td><td>TEST</td><td>111111-11-1111</td><td>Female</td><td>12</td><td>RM 1,212.00</td><td>Pending</td><td> </td></tr> <tr> <td>2</td><td>TEO KAH WEE</td><td>880011-01-0129</td><td>Male</td><td>Doctor</td><td>RM 12,000.00</td><td>Pending</td><td> </td></tr> </table> <div>Menunjukkan 1-9 daripada 9 rekod</div> <div>1</div> |             |                |         |         |              |         |          | No. | Nama | No. K/P | Jantina | Jawatan | Gaji | Status | Tindakan | 1 | TEST | 111111-11-1111 | Female | 12 | RM 1,212.00 | Pending |  | 2 | TEO KAH WEE | 880011-01-0129 | Male | Doctor | RM 12,000.00 | Pending |  |
| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Nama        | No. K/P        | Jantina | Jawatan | Gaji         | Status  | Tindakan |     |      |         |         |         |      |        |          |   |      |                |        |    |             |         |  |   |             |                |      |        |              |         |  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | TEST        | 111111-11-1111 | Female  | 12      | RM 1,212.00  | Pending |          |     |      |         |         |         |      |        |          |   |      |                |        |    |             |         |  |   |             |                |      |        |              |         |  |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | TEO KAH WEE | 880011-01-0129 | Male    | Doctor  | RM 12,000.00 | Pending |          |     |      |         |         |         |      |        |          |   |      |                |        |    |             |         |  |   |             |                |      |        |              |         |  |

Figure 3.2.8

Figure 3.2.8 shows the list of members with status ‘Pending’. Staffs can approve the application by clicking the ‘Tick’ icon button while reject the application by clicking the ‘Cross’ icon button. Next, the staffs can view the details of the member’s information by clicking the ‘Eye’ icon button.

| <div>KADA System</div> <div>Tentang Kami</div> <div>Perkhidmatan</div> <div>Hubungi Kami</div> <div>Welcome, dhesh</div> <div>Keluar</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |                |         |             |             |        |          |     |      |         |         |         |      |        |          |   |                 |                |      |          |             |      |  |   |             |                |        |           |             |      |  |   |             |                |      |    |             |      |  |   |   |                |      |             |             |      |  |   |             |                |        |    |             |      |  |   |         |                |      |            |           |      |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|---------|-------------|-------------|--------|----------|-----|------|---------|---------|---------|------|--------|----------|---|-----------------|----------------|------|----------|-------------|------|--|---|-------------|----------------|--------|-----------|-------------|------|--|---|-------------|----------------|------|----|-------------|------|--|---|---|----------------|------|-------------|-------------|------|--|---|-------------|----------------|--------|----|-------------|------|--|---|---------|----------------|------|------------|-----------|------|--|
| <div>Senarai Ahli</div> <div>Export</div> <div> <div>9 Jumlah Ahli</div> <div>2 Pending</div> <div>6 Ahli Aktif</div> <div>1 Ditolak</div> </div> <div> <div>Cari ahli...</div> <div>Ahli</div> </div> <table> <tr> <th>No.</th><th>Nama</th><th>No. K/P</th><th>Jantina</th><th>Jawatan</th><th>Gaji</th><th>Status</th><th>Tindakan</th></tr> <tr> <td>1</td><td>WEE LEONG CHUAN</td><td>990000-00-0000</td><td>Male</td><td>Pengguna</td><td>RM 2,000.00</td><td>Ahli</td><td></td></tr> <tr> <td>2</td><td>TAN YEE BAL</td><td>001020-10-1203</td><td>Female</td><td>PENSYARAH</td><td>RM 1,000.00</td><td>Ahli</td><td></td></tr> <tr> <td>3</td><td>MUHAMMAD ZI</td><td>040000-00-0000</td><td>Male</td><td>IT</td><td>RM 9,000.00</td><td>Ahli</td><td></td></tr> <tr> <td>4</td><td>A</td><td>777777-77-7777</td><td>Male</td><td>Pelajar UTM</td><td>RM 4,444.00</td><td>Ahli</td><td></td></tr> <tr> <td>5</td><td>LIM ZI MING</td><td>888888-00-0000</td><td>Female</td><td>IT</td><td>RM 8,000.00</td><td>Ahli</td><td></td></tr> <tr> <td>6</td><td>CHEW CX</td><td>222222-22-2222</td><td>Male</td><td>Free Rider</td><td>RM 300.00</td><td>Ahli</td><td></td></tr> </table> |                 |                |         |             |             |        |          | No. | Nama | No. K/P | Jantina | Jawatan | Gaji | Status | Tindakan | 1 | WEE LEONG CHUAN | 990000-00-0000 | Male | Pengguna | RM 2,000.00 | Ahli |  | 2 | TAN YEE BAL | 001020-10-1203 | Female | PENSYARAH | RM 1,000.00 | Ahli |  | 3 | MUHAMMAD ZI | 040000-00-0000 | Male | IT | RM 9,000.00 | Ahli |  | 4 | A | 777777-77-7777 | Male | Pelajar UTM | RM 4,444.00 | Ahli |  | 5 | LIM ZI MING | 888888-00-0000 | Female | IT | RM 8,000.00 | Ahli |  | 6 | CHEW CX | 222222-22-2222 | Male | Free Rider | RM 300.00 | Ahli |  |
| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Nama            | No. K/P        | Jantina | Jawatan     | Gaji        | Status | Tindakan |     |      |         |         |         |      |        |          |   |                 |                |      |          |             |      |  |   |             |                |        |           |             |      |  |   |             |                |      |    |             |      |  |   |   |                |      |             |             |      |  |   |             |                |        |    |             |      |  |   |         |                |      |            |           |      |  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | WEE LEONG CHUAN | 990000-00-0000 | Male    | Pengguna    | RM 2,000.00 | Ahli   |          |     |      |         |         |         |      |        |          |   |                 |                |      |          |             |      |  |   |             |                |        |           |             |      |  |   |             |                |      |    |             |      |  |   |   |                |      |             |             |      |  |   |             |                |        |    |             |      |  |   |         |                |      |            |           |      |  |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | TAN YEE BAL     | 001020-10-1203 | Female  | PENSYARAH   | RM 1,000.00 | Ahli   |          |     |      |         |         |         |      |        |          |   |                 |                |      |          |             |      |  |   |             |                |        |           |             |      |  |   |             |                |      |    |             |      |  |   |   |                |      |             |             |      |  |   |             |                |        |    |             |      |  |   |         |                |      |            |           |      |  |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MUHAMMAD ZI     | 040000-00-0000 | Male    | IT          | RM 9,000.00 | Ahli   |          |     |      |         |         |         |      |        |          |   |                 |                |      |          |             |      |  |   |             |                |        |           |             |      |  |   |             |                |      |    |             |      |  |   |   |                |      |             |             |      |  |   |             |                |        |    |             |      |  |   |         |                |      |            |           |      |  |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | A               | 777777-77-7777 | Male    | Pelajar UTM | RM 4,444.00 | Ahli   |          |     |      |         |         |         |      |        |          |   |                 |                |      |          |             |      |  |   |             |                |        |           |             |      |  |   |             |                |      |    |             |      |  |   |   |                |      |             |             |      |  |   |             |                |        |    |             |      |  |   |         |                |      |            |           |      |  |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | LIM ZI MING     | 888888-00-0000 | Female  | IT          | RM 8,000.00 | Ahli   |          |     |      |         |         |         |      |        |          |   |                 |                |      |          |             |      |  |   |             |                |        |           |             |      |  |   |             |                |      |    |             |      |  |   |   |                |      |             |             |      |  |   |             |                |        |    |             |      |  |   |         |                |      |            |           |      |  |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CHEW CX         | 222222-22-2222 | Male    | Free Rider  | RM 300.00   | Ahli   |          |     |      |         |         |         |      |        |          |   |                 |                |      |          |             |      |  |   |             |                |        |           |             |      |  |   |             |                |      |    |             |      |  |   |   |                |      |             |             |      |  |   |             |                |        |    |             |      |  |   |         |                |      |            |           |      |  |

Figure 3.2.9

Figure 3.2.9 shows the list of members that have been approved with status ‘Ahli’.

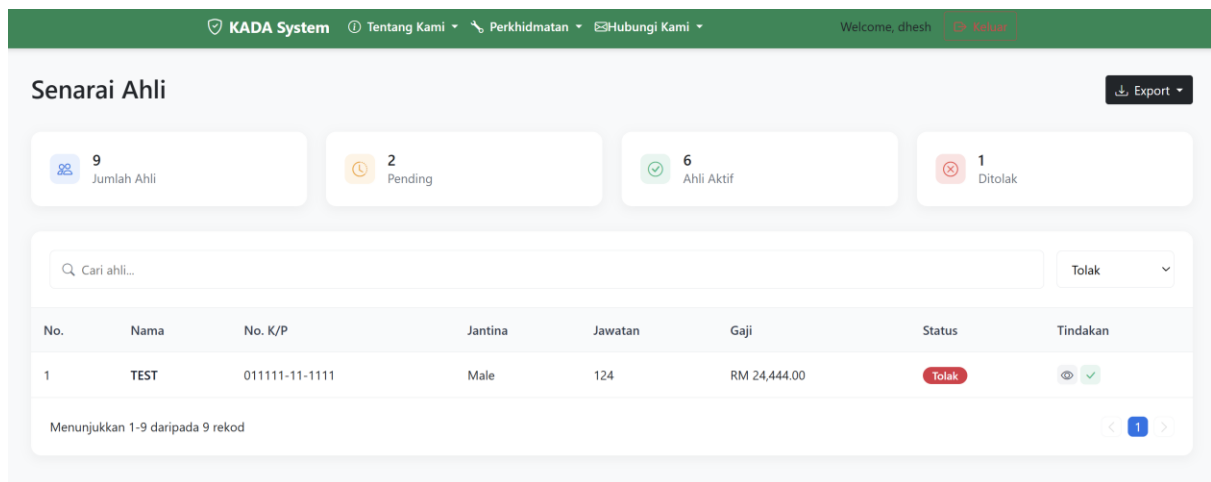


Figure 3.2.10

Figure 3.2.10 shows the list of members that have been rejected with status ‘Tolak’.

## **SECTION D (CONCLUSION AND REFLECTION)**

### **4.1 CONCLUSION**

The Member Registration and Management module has successfully improved the registration process of the cooperative, changing it from a formerly manual system to a more automated and user-friendly platform. This achievement has increased the productivity by allowing more members to register online through an easy process with less need for physical presence. The module also revolutionizes administrative processes in the direction of better validation workflow, allowing staff to organize the handling of applications through a task queue that prioritizes received applications based on their time stamps. Transparency for applicants has been improved with features allowing users to check their application status (e.g., Pending, Approved, or Rejected), thereby boosting user satisfaction and confidence in the system.

Despite these improvements, there is still room for future enhancements to further optimize the system. Implementing an automated validation method, such as rule-based or AI-powered validation, may reduce staff workloads and speed up the validation process. Enhanced notification tools, including specific reasons for rejection and SMS/email alerting, would enable candidates to be more responsive. Besides, the administrators could also use analytics and reporting tools to obtain insights into application behaviors and improve workflows. Extending the platform to a mobile version and incorporating the newest security features, such as multi-factor authentication and data encryption, will ensure continued accessibility while keeping sensitive information secure. These enhancements will ultimately create a more resilient and scalable system that aligned with the cooperative's long-term goals.

## 4.2 REFLECTION

### **Description (Pure fact)**

The Member Registration and Management module was developed as part of the group's effort to create the Koperasi KADA All-in-One Financial Platform. The project aimed to improve the cooperative's operations, replacing manual workflows with a digital system. My assigned task was to design and implement the Member Registration module, focusing on enabling seamless online registration and streamlining the application validation process. The team included members responsible for modules such as login, savings, loans, and reporting. While I focused on database design and implementing SQL queries for my own module, other team members worked on their own modules by developing complementary features like reporting and user interfaces. Collaborative discussions and regular updates helped ensure alignment and smooth progress.

### **Description (Feelings)**

At first, I was somewhat apprehensive yet excited as this is a significant project requiring both technical and collaborative skills. The challenges came along while designing the database schema and ensuring that it was in accordance with the normalization standard, particularly during the alignment with other team members' modules. Frustrations were expressed while debugging SQL queries or getting unexpected data integrity errors, but this was countered with relief after successfully working through certain solutions. Besides, the collaboration spirit formed among the team and effective communication were needed to ease the process. However, resource availability was an issue as balancing the time for testing and modifying the module together with other academic tasks was tough.

### **Evaluation**

The project provided plenty of benefits such as the ability to put theoretical knowledge to practical use and experience visible outcomes of my work. The structured database design and validation workflow built up the system's efficiency and functionality, which was beneficial. On the other side, debugging and optimizing SQL queries under tight deadlines were too stressful. Those moments occasionally interfered with my ability to focus and behave, making it harder to communicate effectively with my team members. The feedback from peers and the lecturer helped to improve the module, but sometimes coordination with each other introduced delays due to different levels of skills and progress rates.

### **Analysis**

I chose the experience of debugging SQL queries and ensuring database integrity during the development of the Member Registration and Management module. The problems of improperly archived rejected applications highlighted the need for more extensive testing and frequent database structure validation is very important. This experience has taught me the value of a more systematic approach, focusing on communication with the team to identify and resolve inter-module dependencies issues. To provide a smoother workflow, future projects should include automated testing and validation scripts to identify similar issues early on.

## **Conclusion**

This project taught me how to balance individual contributions with team coordination. While addressing complex technical challenges ranging from database normalization to debugging and optimizing SQL queries, I learned to address such problems step by step. Furthermore, I realized that continuous communication is beneficial to set up expectations and resolve issues in their early stages. Some concerns remain about how to improve the validation process and increase scalability for larger datasets which I plan to address these points in future projects.

## **Self-criticism**

To enhance my performance in future projects, I will improve my time management skills by creating a formal task plan with plenty of time for testing and integration. I will elicit more immediate feedback from my team members and the lecturer in order to identify possible problems at an earlier stage. Moreover, I plan to use more advanced debugging tools and even try to automate some parts of the validation process to save time. In addition, improving my knowledge of advanced SQL and optimization strategies will provide a foundation for addressing any future similar challenges. All of these steps will ensure more efficient and high quality results in the future projects.